

Xinran Zhang



EDUCATION BACKGROUND

Tsinghua University, Beijing, China

Sept. 2024 – Jun. 2027 (expected)

- Master of Mechanical Engineering | Mechanical Engineering: A+ in MOE National Discipline Assessment
- GPA: 4.0 / 4.0 (Rank: **1/74**)
- Honors: University First-Class Scholarship (2025)

Hunan University, Changsha, China

Sept. 2020 – Jun. 2024

- Bachelor of Vehicle Engineering | Mechanical Engineering: A- in MOE National Discipline Assessment
- GPA: 3.91 / 4.0 (Rank: **1/123** for three consecutive academic years)
- Honors: National Scholarship (2021, 2022, 2023), Outstanding Graduate, Exemplary Student Award

RESEARCH INTERESTS

Physical signal-based model for industrial and robotic systems: multimodal basic model for physical signals such as vibration, acoustics, and haptics; physical-signal-based large models in robot tactile perception / embodied intelligence; industrial intelligent agent.

PUBLICATIONS AND MANUSCRIPTS

Xinran Zhang, Jinfeng Huang, et al. VibAlign: Learning Language-Aligned Signal Tokens for Vibration Signal Analysis with Large Language Models. *Submitted to Mechanical Systems and Signal Processing*.

Xinran Zhang, Qi Li, et al. IADG-BiMamba: Imbalance-Aware Domain Generalization with Bidirectional Mamba for Cross-Condition Bearing Fault Diagnosis. *Submitted to IEEE Sensors Journal*.

Xinran Zhang, Jinfeng Huang, et al. FaultOvis: A Domain-Generalized Vision-Language Model for Rolling Bearing Fault Diagnosis. *Submitted to IEEE Transactions on Instrumentation And Measurement*.

Qi Li[†], **Xinran Zhang**[†], et al. VSLLaVA: A pipeline for Large Multimodal Foundation Models in industrial vibration signal analysis. *Advanced Engineering Informatics*, 2026, 76: 105023.

[†]Equal contribution.

RESEARCH EXPERIENCE

Physical Signal Foundation Models for Industrial and Robotic Systems

Nov. 2025 – Present

Master's Thesis Project, Tsinghua University

- Investigate how LLMs and MLLMs can be adapted to non-text physical signals, using vibration and robotic tactile data as testbeds for sensor-based perception, cross-modal alignment, and natural-language reasoning.
- Advance a progressive research pipeline from signal-image-based visual instruction tuning to domain-generalized VLM representation learning and raw 1-D signal tokenization, leading to three paper-level subprojects on physical-signal foundation model adaptation.

Raw Physical Signal Tokenization and Language Alignment for LLM

Jun. 2026 – Present

Graduate Research Project, Tsinghua University

- Developed VibAlign, a raw physical-signal-to-LLM framework that maps 1-D vibration segments from industrial and robotic tactile sensing scenarios into continuous LLM-compatible signal-token embeddings.
- Designed a BiMamba-based temporal signal encoder, a learnable soft codebook connector, and a three-stage signal-language alignment pipeline combining encoder pretraining, signal-text contrastive alignment, and LoRA-based LLM adaptation with verbalizer-guided supervision.
- Evaluated VibAlign on robotic tactile and industrial vibration benchmarks, achieving **91.84%/91.34%** Acc./F1 on tactile material recognition and **98.18%/98.17%** average Acc./F1 on public industrial vibration benchmarks, with further validations on candidate scoring, signal-embedding utilization, ablations, and cross-dataset generalization.

Domain-Generalized VLM Adaptation for Industrial Sensor Images

Oct. 2025 – Apr. 2026

Graduate Research Project, Tsinghua University

- Developed FaultOvis, an Ovis2-1B-based vision-language adaptation framework that uses STFT images as physical-signal visual inputs for domain-shifted industrial signal reasoning.
- Introduced visual-side discriminative supervision with an auxiliary classification head, a contrastive learning head, and a two-stage training strategy to improve category separability and domain-invariant visual representation learning.
- Evaluated FaultOvis on CWRU and Ottawa leave-one-domain-out benchmarks, achieving the highest Acc./F1 on 7 of 8 unseen-domain tasks and up to **99.33%/99.00%** Acc./F1, with F1 improvements of up to **+76.05** percentage points over the base Ovis2-1B.

Instruction Tuning and RL Refinement for Physical-Signal LMMs

Apr. 2025 – Sep. 2025

Graduate Research Project, Tsinghua University

- Developed VSLLaVA, a large multimodal model adaptation pipeline that formulates physical signal analysis as visual question answering for signal type identification, parameter reasoning, and explanatory responses.
- Constructed an expert knowledge-guided Signal-Question-Answer dataset from simulated and real vibration signals, encoding signal characteristics, physical parameters, frequency-domain patterns, and domain knowledge into multimodal instruction-tuning data.
- Fine-tuned six VLM backbones with LoRA-based SQA-SFT, improving Ovis2-8B Word Recall from 64.29% to 80.81% and CIDEr from 0.16 to 5.52; GRPO further improved InternVL3-8B signal-type identification Acc./F1 from 79.36%/76.99% to 97.50%/97.48%.

COMPETITION AWARDS

Honorable Mention , Mathematical Contest in Modeling (MCM/ICM)	2023
National First Prize , National English Competition for College Students (NECCS)	2023
National Third Prize , The 14th Chinese Mathematics Competition for College Students	2023
Second Prize , Hunan Provincial Mechanics Competition	2022

TEACHING & LEADERSHIP EXPERIENCES

Teaching Assistant

2024 – 2025

Tsinghua University

- Supported *Fundamentals of Engineering Drawing* through SolidWorks labs, grading, Q&A, and proctoring for undergraduates; rated **Excellent** in the final TA evaluation.

Leader, Peer Learning Support Group

2021 – 2024

Hunan University

- Led peer study and exam-preparation activities for classmates; received the **Star of Academic Support** award.

TECHNICAL SKILLS AND LANGUAGE

Programming	Python, MATLAB, C/C++, LaTeX
Deep Learning	PyTorch, Hugging Face Transformers, LLM/VLM adaptation, LoRA/PEFT, multimodal instruction tuning, GRPO/RL refinement, contrastive learning, representation learning
Signal Processing	Vibration and tactile signal preprocessing, time-frequency analysis, short-time Fourier transform, envelope spectrum analysis, feature extraction and visualization
Tools	Linux, Git, VS Code, Origin, Visio, AutoCAD, SolidWorks
English	TOEFL iBT 5.5/6.0, equivalent to 110 on the previous 0–120 scale